

Lab 4: Flood fill and Boundary fill Algorithm

LAB SHEET

Theory: Flood fill Vs Boundary Fill algorithm

Lab works:

Question 1: Write a code in C to implement 4-connected flood fill algorithm.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<graphics.h>
```

```
#include<dos.h>
```

```
void flood(int,int,int,int);
```

```
int main()
```

```
{
```

```
    int gd=DETECT,gm;
```

```
    initgraph(&gd,&gm,"NULL");
```

```
    rectangle(50,50,250,250);
```

```
    flood(55,55,10,0);
```

```
    getch();
```

```
        return 0;
```

```
}
```

```
void flood(int x,int y,int fillColor, int defaultColor)
```

```
{
```

```
    if(getpixel(x,y)==defaultColor)
```

```
    {
```

```

    delay(1);
    putpixel(x,y,fillColor);
    flood(x+1,y,fillColor,defaultColor);
    flood(x-1,y,fillColor,defaultColor);
    flood(x,y+1,fillColor,defaultColor);
    flood(x,y-1,fillColor,defaultColor);
}
}

```

Question 2: Write a code in C to implement boundary fill algorithm.

```

#include<stdio.h>
#include<graphics.h>
#include<dos.h>

void boundaryfill(int x,int y,int f_color,int b_color)
{
    if(getpixel(x,y)!=b_color && getpixel(x,y)!=f_color)
    {
        putpixel(x,y,f_color);
        boundaryfill(x+1,y,f_color,b_color);
        boundaryfill(x,y+1,f_color,b_color);
        boundaryfill(x-1,y,f_color,b_color);
        boundaryfill(x,y-1,f_color,b_color);
    }
}

```

```

int main()
{
    int gm,gd=DETECT,radius;
    int x,y;
    printf("Enter x and y positions for circle\n");
    scanf("%d%d",&x,&y);
    printf("Enter radius of circle\n");
    scanf("%d",&radius);
    initgraph(&gd,&gm,"NULL");
    circle(x,y,radius);
    boundaryfill(x,y,4,15);
    delay(5000);
    closegraph();
    return 0;
}

```

Conclusion

- What did u learn?